

STM32F103C8 Architecture & Low-Power Mode

Device: **STM32F103C8** Cortex-M3 core, up to 72 MHz, 64 KB Flash, 20 KB SRAM.

Primary sources

- **RM0008** — *Reference manual: STM32F101/102/103/105/107xx family*, Rev 21, Feb 2021 (ST DocID: RM0008)
- **DS5319** — *STM32F103x8/STM32F103xB datasheet*, Rev 20, Jul 2025 (ST DocID: DS5319)
- **PM0056** — *STM32F10xxx/20xxx/21xxx/L1xxxx Cortex®-M3 programming manual*, Rev 7, Dec 2024 (ST DocID: PM0056)

mostly reference manual is used if others their name are used

1. STM32 Architecture Overview

Cortex-M3 Core

- 32-bit ARMv7-M architecture, 3-stage pipeline, Thumb-2 instruction set (, p.11).
- Built-in **NVIC** (Nested Vectored Interrupt Controller), handling up to 43 maskable interrupt channels plus 16 core exceptions (p.197).
- Built-in **SysTick** timer — 24-bit downcounter (p.150).
- **SCB** (System Control Block) is the programmer's-model interface to the core, including the **SCR** used to configure sleep behavior (p.136).
- No FPU on Cortex-M3 (that arrives with Cortex-M4F).

Clock System (RCC – Reset and Clock Control)

Register block described in **RM0008 Chapter 7, p.90**.

Source	Frequency	Register	Citation
HSI	8 MHz internal RC, default at reset	RCC_CR	p.99
HSE	4–16 MHz external crystal	RCC_CR	p.99
PLL	Multiplies HSI/2 or HSE up to 72 MHz	RCC_CFGR	p.101
LSE	32.768 kHz for RTC	RCC_BDCR	p.118
LSI	~40 kHz for IWDG/RTC	RCC_CSR	p.119

(Bus prescalers: **AHB (HCLK)** up to 72 MHz, **APB1 (PCLK1)** max 36 MHz, **APB2 (PCLK2)** max 72 MHz p.92). A peripheral's clock must be individually enabled via `RCC_AHBENR` ,

RCC_APB1ENR (RM0008 §7.3.8, p.115), or RCC_APB2ENR (p.112) before its registers are accessible — this includes PWREN for the Power Controller.

GPIO

- Ports A–E on this device family, described in **RM0008 Chapter 9, p.159**.
- Configuration via GPIOx_CRL (pins 0–7, p.171) and GPIOx_CRH (pins 8–15, , p.172).
- Data via GPIOx_IDR (p.172), GPIOx_ODR (p.173), atomic set/reset via GPIOx_BSRR (p.173) / GPIOx_BRR (p.174).
- GPIO pins feed the EXTI controller through the AFIO external-interrupt configuration registers AFIO_EXTICR1..4 (pp.191–192).

Interrupt Controller (NVIC)

- Up to 43 maskable interrupt channels for this family, plus 16 core exceptions, and 16 priority levels (DS5319 §2.3.5, p.14).
- Registers: NVIC_ISERx (PM0056 §4.3.2, p.120), NVIC_ICERx (PM0056 §4.3.3, p.121), NVIC_ISPRx (PM0056 §4.3.4, p.122), NVIC_ICPRx (PM0056 §4.3.5, p.123), NVIC_IPRx priority registers (PM0056 §4.3.7, p.125), full map at PM0056 §4.3.11, p.128.

Peripheral Architecture

- Memory map places PWR at 0x4000 7000 , RCC at 0x4002 1000 , EXTI at 0x4001 0400 , AFIO at 0x4001 0000 (p.50).
- PWR and BKP (backup) domains stay powered from VDD / VBAT across most low-power modes (p.69).

Power-Control System

- **PWR block** manages regulator state, low-power entry conditions, wake-up pin, and status flags (p.67).
- Two power domains: **VDD domain** (core logic, most peripherals — shut down in Standby) and **Backup domain** (VBAT : RTC, LSE oscillator, backup registers — survives Standby and VDD loss) (p.69).
- Regulator has three states: **Main/Run**, **Low-power** (used in Stop), and **Power-down** (Standby) (p.16).

2. STM32 Low-Power Modes (STM32F103C8)

The STM32F103 mainstream line implements **Sleep, Stop, and Standby** only (p.16). **"Low-power Sleep"** and **"Shutdown"** are named modes on later ultra-low-power families (STM32L0/L1/L4/L5, G0/G4, U5) and do not apply to the F103.

Mode	Core clock	Peripheral clocks	SRAM/Registers	Regulator	Citation
Sleep	Stopped	Running	Retained	Run	p.73
Stop	Stopped	Stopped (HSI/HSE/PLL off)	Retained	Run or Low-power	p.74
Standby	Off	Off	Lost (except backup domain)	Off	p.76

- **Sleep**: only the Cortex-M3 core clock is gated; peripherals continue running and can wake the core (p.73, p.74).
- **Stop**: all 1.8 V-domain clocks stop (HSI, HSE, PLL disabled); SRAM/registers preserved; wake-up is via any EXTI line (p.74).
- **Standby**: 1.8 V domain fully powered off; SRAM/registers lost except backup domain and standby circuitry; wake is equivalent to a power-on reset (p.76).
- **Auto-wakeup (AWU)** from low-power mode using the RTC alarm is described separately (p.77).

Measured current consumption (DS5319)

Mode	Typical current	Citation
Sleep (max, all peripherals enabled, 72 MHz)	30 mA (TA=85 °C)	DS5319 Table 15, p.43
Stop, regulator in Run mode	~23.5–24 µA typ	DS5319 Table 16, p.44
Stop, regulator in Low-power mode	~13.5–14 µA typ	DS5319 Table 16, p.44
Standby, LSI + IWDG off, LSE/RTC off	~1.7–2 µA typ	DS5319 Table 16, p.44
Wake-up latency: Sleep / Stop (Run reg.) / Stop (LP reg.) / Standby	1.8 µs / 3.6 µs / 5.4 µs / 50 µs	DS5319 Table 26, p.56

3. Important Registers for Low-Power Operation

PWR registers — base 0x4000 7000 (RM0008 §5.4, p.77; memory map DS5319 Fig.11, p.34)

Register	Purpose	Citation
PWR_CR	Regulator/mode select: LPDS , PDDS , CWUF , CSBF , PVD bits	p.77
PWR_CSR	Wake flags (WUF , SBF) and EWUP (WKUP pin enable)	p.79
PWR register map	Full bit map and reset values	p.80

PWREN in RCC_APB1ENR (RM0008 §7.3.8, p.115) must be set before PWR_CR / PWR_CSR are accessible.

RCC registers — base 0x4002 1000 (p.90)

Register	Purpose	Citation
RCC_CR	HSI/HSE/PLL enable & ready flags	p.99
RCC_CFGR	System clock switch/status, prescalers	p.101
RCC_APB1ENR	PWREN and other APB1 clock gates	p.115
RCC_APB2ENR	GPIO/AFIO clock gates	p.112
RCC_BDCR	Backup domain: LSEON , RTCSEL , RTCEN	p.118
RCC_CSR	LSION and reset-cause flags	p.119

Cortex-M3 SCB — System Control Register (SCR)

- Address 0xE000ED10 , described in **PM0056 §4.4.6, p.136**.
- SLEEPONEXIT (bit 1), SLEEPDEEP (bit 2), SEVONPEND (bit 4) — all in the same register (PM0056 §4.4.6, p.136).
- The power-management behavior these bits drive is explained conceptually in **PM0056 §2.5, p.41** ("Power management"), with sleep entry mechanisms at **§2.5.1, p.42** and wake-up conditions at **§2.5.2, p.42**.

EXTI registers — base 0x4001 0400 (RM0008 §10.3, p.211)

Register	Purpose	Citation
EXTI_IMR	Interrupt mask register	p.211
EXTI_EMR	Event mask register (for WFE)	p.211
EXTI_RTSTR	Rising trigger selection	p.212
EXTI_FTSTR	Falling trigger selection	p.212
EXTI_SWIER	Software interrupt/event register	p.213
EXTI_PR	Pending register	p.213

Register	Purpose	Citation
EXTI register map	Full map and reset values	p.214

The F103 has 19 EXTI lines: **EXTI0–15** map to a GPIO pin selected via `AFIO_EXTICR1` . . 4 (RM0008 §9.4.3–9.4.6, pp.191–192); **EXTI16** = PVD, **EXTI17** = RTC Alarm, **EXTI18** = USB wake-up (RM0008 §10.2.5, p.209; DS5319 §2.3.6, p.15 confirms 19 lines and up to 80 GPIOs routable to the 16 external lines).

NVIC / interrupt configuration registers (PM0056 §4.3, p.118)

Register	Purpose	Citation
NVIC_ISERx	Interrupt set-enable	PM0056 §4.3.2, p.120
NVIC_ICERx	Interrupt clear-enable	PM0056 §4.3.3, p.121
NVIC_ISPRx	Set-pending	PM0056 §4.3.4, p.122
NVIC_ICPRx	Clear-pending	PM0056 §4.3.5, p.123
NVIC_IPRx	Priority configuration (4 bits used on F103)	PM0056 §4.3.7, p.125

4. Wake-Up Sources

Source	Stop	Standby	Citation
Any EXTI line (0–15)	Yes	Only via WKUP pin (PA0)	p.74, p.76
RTC Alarm	Yes (EXTI17)	Yes	p.209, p.76
NRST pin	Yes	Yes (full reset)	DS5319 §2.3.12, p.16
IWDG reset	Yes	Yes	DS5319 §2.3.12, p.16
WKUP pin (PA0), enabled via <code>PWR_CSR.EWUP</code>	Yes	Yes (primary Standby source)	p.79
USB wake-up (EXTI18)	Yes	No	p.209

Practical Stop-mode wake sequence: enable `PWREN` (p.115) → configure EXTI line (`AFIO_EXTICRx` ,p.191; `EXTI_IMR` / `EXTI_RTSR` / `FTSR` p.211) → clear `PWR_CR.PDDS` (p.77) → set `SCB->SCR.SLEEPDEEP` (PM0056 §4.4.6, p.136) → execute `WFI` / `WFE` (PM0056 §3.9.11–3.9.12, pp.103–104).

Practical Standby wake sequence: set `PWR_CR.CWUF` and `PWR_CSR.EWUP` (pp.77–79) → set `PWR_CR.PDDS=1` and `SLEEPDEEP=1` → WFI → check `PWR_CSR.SBF` at boot (p.79).

5. WFI and WFE (Cortex-M3)

Both are documented in **PM0056 Chapter 3 "The Cortex-M3 instruction set," §3.9, p.97**, and their role in entering low-power modes is explained conceptually in **PM0056 §2.5 "Power management," p.41**.

WFI — Wait For Interrupt

- Instruction description: **PM0056 §3.9.12, p.104**.
- Wakes on any interrupt enabled at both the NVIC (PM0056 §4.3.2, p.120) and, where relevant, the EXTI mask (p.211).
- Wake-up behavior for WFI/sleep-on-exit is detailed in **PM0056 §2.5.2 "Wakeup from sleep mode," p.42**.

WFE — Wait For Event

- Instruction description: **PM0056 §3.9.11, p.103**.
- Event sources include EXTI lines armed via `EXTI_EMR` (p.211), or any pending interrupt if `SEVONPEND` is set in `SCB_SCR` (PM0056 §4.4.6, p.136).
- The external event input mechanism (EXTI feeding the core's event line) is described in **PM0056 §2.5.3, p.43**.
- Programming hints (CMSIS `__WFE()` / `__WFI()` intrinsics) are given in **PM0056 §2.5.4, p.43**.

Mode-entry quick reference

Target mode	<code>PWR_CR.PDDS</code>	<code>SCB->SCR.SLEEPDEEP</code>	Instruction	Citation
Sleep	—	0	WFI / WFE	p.73
Stop	0	1	WFI / WFE	p.74, p.77
Standby	1	1	WFI	p.76,p.77